

6.2

$$B = \left(\left(\overset{\vec{u}_1}{\begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}}, \begin{pmatrix} \overset{\vec{u}_2}{1} \\ 0 \\ 1 \\ 2 \end{pmatrix} \right) \right)$$

$$\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} = \vec{v}$$

$$\left(\begin{array}{cc|ccc} 2 & 1 & 4 & 2 & 5 \\ 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & -2 & 3 \\ 1 & 2 & 5 & -2 & 7 \end{array} \right) \sim \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & -2 & 3 \\ 2 & 1 & 4 & 2 & 5 \\ 1 & 2 & 5 & -2 & 7 \end{array} \right) \sim$$

$$\sim \left(\begin{array}{cc|ccc} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & -2 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$[id]_B^{K_2}$$

 $\downarrow I_2$

$$f(\vec{v}) = [id]_B^{K_2} [f]_{K_2}^{K_2} [id]_B^{K_2} [\vec{v}]_B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

ii
h

$$h(\vec{v}) = \underline{\begin{pmatrix} 1 \\ 3 \end{pmatrix}} \quad \begin{vmatrix} 1-x & 2 \\ 2 & -2-x \end{vmatrix} = x^2 + x - 6 = (x+3)(x-2)$$

v.l. c. -3 a 2

$$M_{-3} = \text{Span}\left\{ \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\} \Leftarrow \left(\begin{array}{cc|c} 4 & 2 & 0 \\ 2 & 1 & 0 \end{array} \right)$$

$$M_2 = \text{Span}\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\} \Leftarrow \left(\begin{array}{cc|c} -1 & 2 & 0 \\ 2 & -4 & 0 \end{array} \right)$$

$$h(x) = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix} \frac{1}{5} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$

$$[f^n]_{\mathcal{B}}^{\mathcal{B}} \vec{v} = [id]_{\mathcal{B}}^{k_2} [f^n]_{\mathcal{B}}^{k_2} [\vec{v}]_{\mathcal{B}} = *$$

$\parallel$
 $h^n(x)$

$\parallel$

$$\frac{1}{5} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$$

$$* \frac{1}{5} \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} =$$

$$= \frac{1}{5} \begin{pmatrix} 5 & 0 \\ 2 & 1 \\ 1 & -2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 5 \\ -5 \end{pmatrix} =$$

$$= \begin{pmatrix} 5 & 0 \\ 2 & 1 \\ 1 & -2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2^n & 0 \\ 0 & (-3)^n \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} =$$

$$= \begin{pmatrix} 5 & 0 \\ 2 & 1 \\ 1 & -2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2^n \\ -(-3)^n \end{pmatrix} =$$

$$= \begin{pmatrix} 5 \cdot 2^n \\ 2^{n+1} - (-3)^n \\ 2^n + 2 \cdot (-3)^n \\ 2^n - 2 \cdot (-3)^n \end{pmatrix} = f^n \left(\begin{pmatrix} 5 \\ 1 \\ 3 \\ 7 \end{pmatrix} \right)$$
